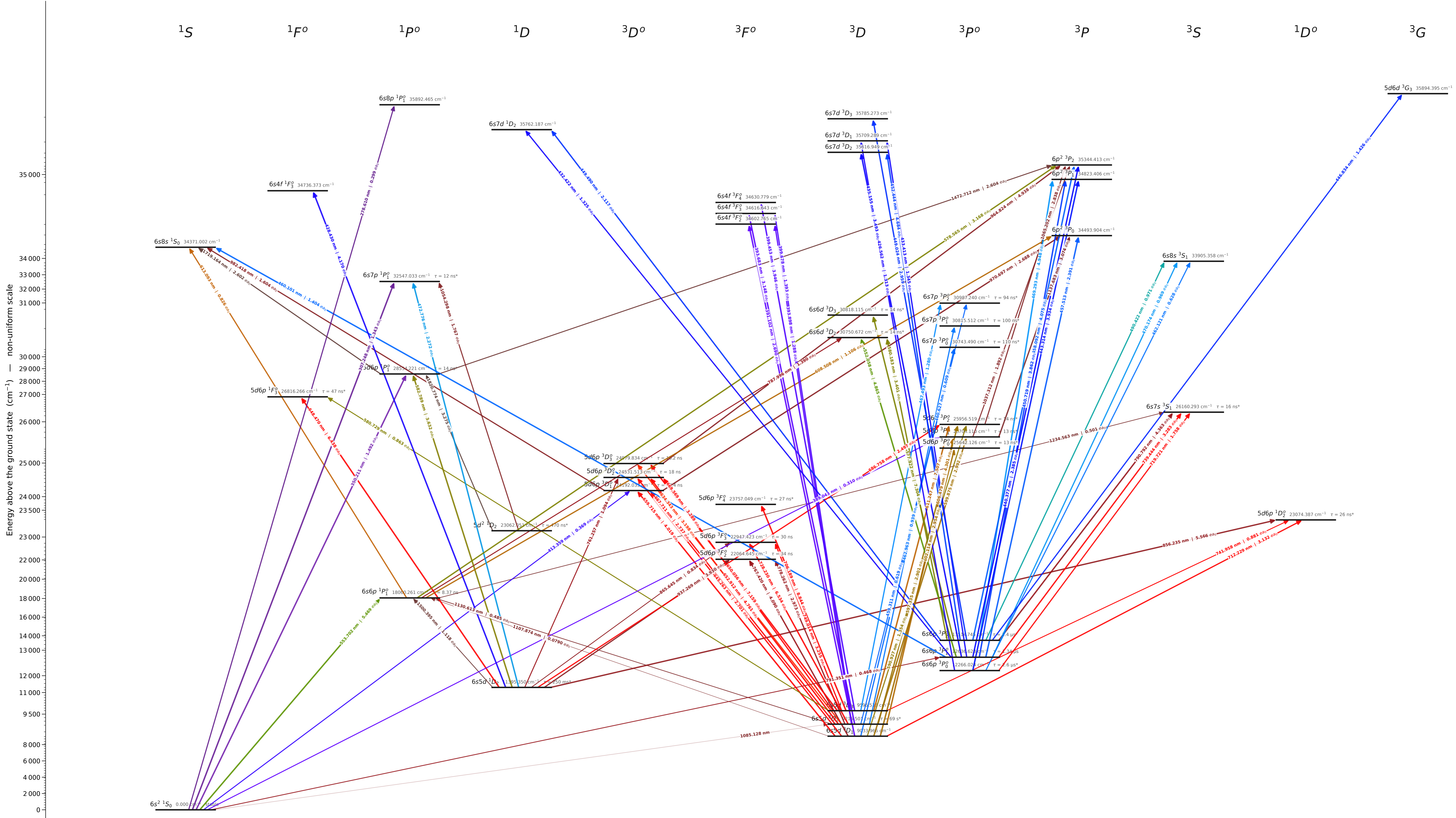
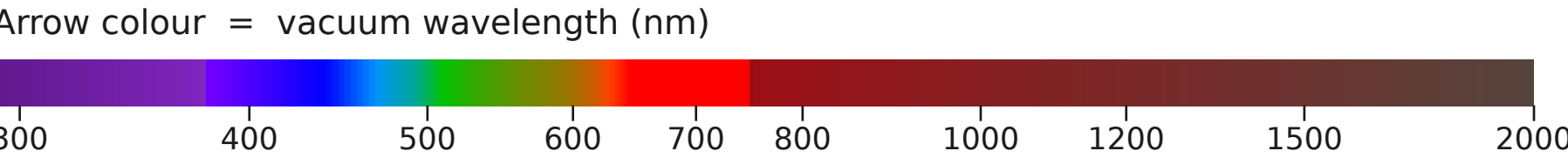


Barium-137

¹³⁷Ba (Ba I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 44 levels, 89 lines.

Ba ⁺ ionisation limit 42034.91 cm⁻¹ = 5.21166 eV (237.897 nm)

Arrow colour = vacuum wavelength | reduced dipole matrix element, $|U||er||U|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er U' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6s ² 1S ₀ - 5d6p 3D ₂ ^o	413.3592	413.2427	725.2589	0.369	1.300e+06	9.147	2.3	413.36 nm intercombination line: fluorescence detection o...
6s ² 1S ₀ - 6s6p 1P ₁ ^o	553.7019	553.5481	541.4330	5.469	1.190e+08	19.01	99.7	main cooling / imaging line
6s5d 3D ₁ - 5d6p 3D ₂ ^o	659.7147	659.5326	454.4274	4.019	3.800e+07	9.147	64	659.7 nm repumper
6s5d 3D ₂ - 5d6p 3D ₂ ^o	667.7113	667.5270	448.9851	2.737	1.700e+07	9.147	32	667.7 nm repumper
6s ² 1S ₀ - 6s6p 3P ₁ ^o	791.3507	791.1330	378.8364	0.468	2.990e+05	0.1183	40.2	intercombination line
6s ² 1S ₀ - 6s5d 3D ₂	1085.1282	1084.8310	276.2738	-	1.454e-02	2.307e-09	100	E2 decay limiting the 3D2 lifetime
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.8743	1107.5709	270.6015	0.079	3.100e+03	19.01	0.0026	1107.8 nm decay branch / repumper
6s5d 3D ₂ - 6s6p 1P ₁ ^o	1130.6129	1130.3034	265.1592	0.485	1.100e+05	19.01	0.09	1130.6 nm decay branch / repumper
6s5d 1D ₂ - 6s6p 1P ₁ ^o	1500.3951	1499.9852	199.8090	1.118	2.500e+05	19.01	0.25	1500.4 nm decay branch of the cooling line / repumper

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ(2π): natural linewidth of the upper level, 1/(2πτ).

¹³⁷Ba hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
6s5d 3D ₁	-520.536(3)	17.890(3)	F=1/2: +1323.70 F=3/2: +502.65 F=5/2: -776.33
6s5d 3D ₂	415.928(3)	25.899(13)	F=1/2: -1849.01 F=3/2: -1247.78 F=5/2: -224.15 F=7/2: +1254.26
6s5d 3D ₃	456.559(4)	47.390(16)	F=3/2: -2710.92 F=5/2: -1605.07 F=7/2: -23.70 F=9/2: +2066.36
6s5d 1D ₂	-82.180(3)	59.564(14)	F=1/2: +421.93 F=3/2: +246.54 F=5/2: +3.86 F=7/2: -231.65
6s6p 3P ₁ ^o	1150.590	-41.610	F=1/2: -2928.49 F=3/2: -1108.98 F=5/2: +1715.48
6s6p 1P ₁ ^o	-109.50(13)	50.09(21)	F=1/2: +336.36 F=3/2: +59.41 F=5/2: -151.73
5d6p 3D ₂ ^o	172.9(3)	-3.5(6)	F=1/2: -436.62 F=3/2: -169.40 F=5/2: +258.48

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6s ² 1S ₀ - 5d6p 3D ₂ ^o	413.359	137-138	392.9(9)
6s ² 1S ₀ - 5d6p 3D ₂ ^o	413.359	136-138	178.1(8)
6s ² 1S ₀ - 5d6p 3D ₂ ^o	413.359	135-138	401.4(12)
6s ² 1S ₀ - 5d6p 3D ₂ ^o	413.359	134-138	124.3(13)
6s ² 1S ₀ - 6s6p 1P ₁ ^o	553.702	137-138	215.15(16)
6s ² 1S ₀ - 6s6p 1P ₁ ^o	553.702	136-138	128.0(4)
6s ² 1S ₀ - 6s6p 1P ₁ ^o	553.702	135-138	259.29(17)
6s ² 1S ₀ - 6s6p 1P ₁ ^o	553.702	134-138	143.0(5)
6s5d 3D ₁ - 5d6p 3D ₂ ^o	659.715	137-138	189.2(10)
6s5d 3D ₁ - 5d6p 3D ₂ ^o	659.715	136-138	112.9(10)
6s5d 3D ₁ - 5d6p 3D ₂ ^o	659.715	135-138	228.3(10)
6s5d 3D ₁ - 5d6p 3D ₂ ^o	659.715	134-138	125.0(20)
6s ² 1S ₀ - 6s6p 3P ₁ ^o	791.351	137-138	183.4(10)
6s ² 1S ₀ - 6s6p 3P ₁ ^o	791.351	136-138	109.2(10)
6s ² 1S ₀ - 6s6p 3P ₁ ^o	791.351	135-138	219.9(10)
6s ² 1S ₀ - 6s6p 3P ₁ ^o	791.351	134-138	122.3(25)
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.874	137-138	-114(4)
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.874	136-138	59.3(6)
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.874	135-138	-39(4)
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.874	134-138	144.1(10)
6s5d 3D ₁ - 6s6p 1P ₁ ^o	1107.874	137-135	-75.3(5)
6s5d 3D ₂ - 6s6p 1P ₁ ^o	1130.613	137-138	-69(3)

Shift = ν(first isotope) - ν(second isotope).